

Expand and Exempt: Compensation and the Politics of US Chip Export Controls*

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Abstract

Since October 2022 the United States has widened its export controls on advanced chips and chipmaking equipment, although nearly every commenting firm and trade association asked for less. I ask whether the controls are security statecraft or private commercial interest presented in the language of security. The evidence comes from public comments on 49 federal dockets, scored with a fine-tuned language model validated on 213 hand-coded letters, seven rule revisions and their 57 carve-outs, lobbying disclosures and CHIPS Act awards. The controls are security statecraft that private interest hollows out from within. Chipmakers and their suppliers of tools and materials asked for narrower controls than firms that buy and deploy chips. The Bureau of Industry and Security widened scope in five of six chip-rule revisions over that opposition. Yet those rules reopened flows into China for the firms that resisted most. Chipmaking and equipment firms received 18 of the 24 carve-outs that did so, including all 8 that BIS granted at commenters' request, although their products are the chokepoints behind US leverage over Chinese chipmaking. They also received \$32.8 billion of \$33.5 billion in CHIPS awards, and firms whose awards were finalized lobbied less afterward than firms still waiting.

*Working paper in progress. Please do not cite or circulate.

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1 Introduction

On October 7, 2022, the Bureau of Industry and Security (BIS) at the US Department of Commerce placed license requirements on advanced computing chips, on the computers that contain them and on equipment used to make advanced chips, and it restricted what US persons could do for Chinese fabrication plants. By April 2026 BIS had revised its chip rules six times. Each revision followed a public comment period, and in each period the firms and trade associations that wrote in asked BIS to control less. BIS widened scope in five of the six revisions anyway. Yet the same revisions also reopened flows into China, and those exemptions went mostly to the firms that had resisted hardest, the makers of the equipment and materials that give the United States its hold over Chinese chipmaking.

The sequence poses a question that the debate over US technology policy toward China has not settled. Are chip export controls economic statecraft driven by national security, or are they private commercial interests presented in the language of national security? The security reading has strong support. Advanced chips train frontier AI models, the United States and its allies hold chokepoints in chip design and equipment (Buchanan, 2020; Farrell & Newman, 2019; Sastry et al., 2024), and the October 2022 rule states that BIS sought to restrict China’s “access to advanced computing for its military modernization, including nuclear weapons development” (87 FR 62186). The commercial reading also has support. Security exceptions have long sheltered economic aims in trade law (Heath, 2020; Pinchis-Paulsen, 2020), export controls historically bent to exporters (Mastanduno, 1992),

and the controls arrived alongside \$39 billion in manufacturing subsidies under the CHIPS and Science Act.

I argue that chip export controls are security statecraft that private interest hollows out from within. The state set the direction of the controls, and the firms with the most to lose decided where they would bite. The link between the two is compensation. Controls on chips and chipmaking equipment impose their heaviest costs on firms that make chips and the tools and materials to make them. A state that wants to keep widening such controls has to manage those firms. It can write carve-outs into the rules that soften the controls where those firms are exposed, and it can pay them through industrial policy. Both forms of payment let the state keep expanding scope without the losers turning the rulemaking against it. The carve-outs carry a security cost. The firms that resist most are the ones that hold the chokepoints, so the relief they win reopens part of what the controls were written to close.

The evidence comes from the documentary record of US chip policy between 2022 and 2026. I collect every public comment on 49 federal dockets that bear on export controls, CHIPS incentives and AI policy, 6,641 comments in all, and score each one for whether it asks for wider or narrower control, using a fine-tuned natural language inference model validated out of fold against 213 hand-coded letters. Each commenting organization is placed on the AI compute chain according to what it does. For the seven BIS revisions that followed a commented chip rule, I code whether scope widened, and for the 57 carve-outs those revisions added, who benefits and whether BIS says commenters asked for them. Federal lobbying disclosures are joined to the dates of CHIPS preliminary terms and final awards.

I report three findings. First, firms in fabrication, equipment and materials asked for narrower controls than firms that buy and deploy chips and AI. The gap holds with agency or docket fixed effects and with argument type held fixed. Second, BIS widened scope over unanimous opposition in the comment record, and every revision also added carve-outs. Of the 24 carve-outs that reopened flows into China, 18 reach midstream firms, the coalition that resisted most and whose products give the controls their force. All 8 of the China-flow carve-outs that BIS granted at commenters' request are among them. Midstream items make up only about a third of the controlled provisions in those rules, and across all 57 carve-outs, 34 reach midstream firms. Third, CHIPS money went to the same coalition, \$32.8 billion of \$33.5 billion. Lobbying intensity predicts who received preliminary terms only weakly, and firms whose awards were finalized cut their chip and export-control lobbying relative to firms that received preliminary terms and never finalized. The late expansions against equipment makers and foundries, in December 2024 and January 2025, came in the weeks when most of those firms' awards were finalized.

The paper contributes to three literatures. For work on firms in trade politics (I. S. Kim, [2017](#); I. S. Kim & Osgood, [2019](#); Osgood, [2017](#)), it shows the firm-level cleavage carrying over from tariffs to security controls, with position in a production chain sorting firms within a single industry. For work on exceptions and escape clauses (Pelc, [2016](#); Rosendorff & Milner, [2001](#)) and on business influence in rulemaking (Libgober, [2020a](#); Yackee & Yackee, [2006](#)), it shows carve-outs working as the price of expansion, landing on the coalition that resisted most and reopening flows to the country the controls were aimed at. For work on economic statecraft and the new industrial policy (Baldwin, [1985](#); Bown & Wang, [2024](#); Farrell &

Newman, [2019](#); Juhász & Lane, [2024](#)), it ties subsidies and controls together as one bargain, in which the state paid the losers of a security policy and gave back part of its reach.

Section [2](#) develops the argument and the hypotheses. Section [3](#) describes the rules, the awards and the data. Sections [4](#) through [6](#) report the tests. Section [7](#) returns to the central question, and Section [8](#) concludes with what the record cannot show.

2 Theory

Economic statecraft is the use of economic means for foreign policy ends (Baldwin, [1985](#)). On the security reading, chip controls are statecraft of a familiar kind. The United States and a few allies dominate chip design software, advanced equipment and the most capable chips, and a state that controls the hubs of a network can deny them to rivals (Beaumier & Cartwright, [2024](#); Farrell & Newman, [2019](#)). Compute is one of the inputs that drive AI capability, which makes advanced chips a natural target (Buchanan, [2020](#); Sastry et al., [2024](#)), and by 2025 the United States held about three quarters of the world’s AI compute capacity (Abecassis et al., [2025](#)). Rising powers that threaten an incumbent invite it to weaponize supply chains (Chen & Evers, [2023](#)), and China’s industrial policy in semiconductors posed exactly that threat to US structural power (Malkin, [2022](#)).

The commercial reading starts from the observation that security and economic aims are hard to separate in trade and investment policy (Mastanduno, [1998](#); Roberts et al., [2019](#)). National security claims have expanded and resist review (Heath, [2020](#)), the

GATT security exception was drafted with economic aims in view (Pinchis-Paulsen, 2020), and security screening of foreign investment in Europe tracks national commercial stakes (Chan & Meunier, 2022). Public choice accounts treat sanctions and controls as policies that interest groups shape like any other (Kaempfer & Lowenberg, 1999), and the history of CoCom shows export controls bargained among security agencies, allies and exporters (Mastanduno, 1992). Regulation in general can be acquired by the industry it governs (Stigler, 1971), although claims of capture need evidence that the regulated industry set the content of the rule (Carpenter & Moss, 2013). Securitization theory adds that calling an issue a security matter lifts it out of normal politics (Buzan et al., 1998), which would let commercial rivalry with Chinese firms travel under a security label.

The commercial reading needs US firms that gain from wider controls, and some parts of the chain qualify. Cloud providers and AI developers buy chips, sell little hardware into China and compete with Chinese AI firms. Controls that keep advanced compute from those rivals raise their costs, and control over rivals is one of the main things an industry seeks from regulation (Stigler, 1971). A security label makes such a request easier to defend. US chipmakers that compete with Chinese producers gain in the same way when controls slow those producers down. Firms that want wider controls have several ways into the rules. They can ask for wider scope in public comments (Yackee & Yackee, 2006), press their case in meetings with officials (Libgober, 2020a) and shape drafts before publication, since agencies write proposals with the expected comments in mind (Libgober, 2020b). On this reading, each revision should move toward the coalition that asked for it, and relief should go to the firms with the most influence.

The security reading predicts something different. If security set the controls, the direction should hold against business opposition. Both readings expect the losers to be managed, since a state that pursues a security aim at the expense of domestic firms still has to deal with those firms. They differ on what the relief gives up. On the security reading, carve-outs should trim the controls' reach into domestic and allied operations and leave the flows to the target closed. On the commercial reading, carve-outs should follow the firms with the most at stake, even where they reopen those flows.

2.1 Who Loses from Chip Controls

Firm-level trade politics holds that firms within one industry want different trade policies. Large, productive firms in differentiated industries lobby on their own and for their own products (Bombardini & Trebbi, 2012; I. S. Kim, 2017). Product variety and intra-industry trade break the unified industry opposition that sector models assume (Madeira, 2016; Osgood, 2017). Firms embedded in global supply chains support openness and back trade agreements (Milner, 1988; Osgood, 2018; Zeng et al., 2020), and they hold preferences over several dimensions of trade policy at once (I. S. Kim et al., 2019).

Export controls bring this logic into security policy. A chip control lands on a production chain with three broad stages. Upstream firms design chips, license intellectual property, make design software and do the research the chain depends on. Midstream firms make the equipment and materials for fabrication, run fabs and package and test chips. Downstream firms buy chips or AI services and build them into computers, cloud platforms,

models and applications. The same rule falls on each stage differently.

Controls on chipmaking equipment and on fabrication hit midstream firms hardest, and controls on chips reach them as well, since foundries and packagers make the controlled chips. The controlled items are their products. China is the largest market for the major US equipment makers, with 37 percent of Applied Materials' revenue, 42 percent of Lam Research's and 43 percent of KLA's in fiscal year 2024 (Applied Materials, Inc., [2024](#); KLA Corporation, [2024](#); Lam Research Corporation, [2024](#)), and export controls on Chinese customers impose large losses on their US suppliers (Crosignani et al., [2026](#)). Equipment controls also reach midstream firms' own operations. The October 2022 rule required a license for US persons to service tools in Chinese fabs that make advanced chips (§ 744.6(c)(2)(iii), 87 FR 62199) and for any item subject to the Export Administration Regulations (EAR) that would be used to develop or produce chipmaking equipment or its parts in China (§ 744.23(a)(2)(v), 87 FR 62201). Fabs and memory makers with plants in China face license requirements on their own facilities (§ 744.23(a)(2)(iii), 87 FR 62201). From October 2023, Samsung's and SK hynix's memory plants in China received controlled items under general Validated End-User authorizations (88 FR 71479), which BIS revoked effective December 31, 2025 (90 FR 42321).

Upstream designers lose sales of controlled chips, and those losses can be large. Their exposure, though, depends on where the rule draws its thresholds, and they can redesign products to fall below them. Downstream firms lose least. They buy chips, and controls on sales to China do not raise their input costs at home. Cloud providers and AI developers among them can even gain from controls on their rivals' inputs.

This gives an ordering of opposition. Midstream firms should oppose expansion most, upstream firms should oppose it where thresholds catch their products, and downstream firms should oppose it least. Upstream exposure turns on thresholds that change from rule to rule, so the stable contrast runs between midstream firms and the firms that buy and deploy chips.

H1. Firms in fabrication and its supply base oppose control expansion more strongly than firms that buy and deploy chips.

2.2 How the State Pays and What Payment Buys

A state that widens controls over the objections of midstream firms can pay them in two ways.

The first is regulatory. Rules can be written with exceptions, exclusions and transition periods that soften the control where specific firms are exposed. Trade agreements include escape clauses because flexibility lets governments absorb domestic pressure while keeping the larger commitment (Kucik & Reinhardt, 2008; Rosendorff & Milner, 2001), and how exceptions are designed and who can invoke them is a political choice (Pelc, 2016). Regulatory barriers tend to favor firms large enough to absorb their costs and shape their terms (Gulotty, 2020). Firm-specific relief processes in trade, such as tariff suspensions and exclusions, respond to the information and political connections of the firms that request them (Fotak et al., 2025; Ludema et al., 2018). In export controls the equivalent instruments are license exceptions, exclusion notes on ECCNs, temporary general licenses, savings

clauses and routes by which approved parties can certify their way out of a control. Each is written into the Federal Register text, and each names whom it relieves.

Notice-and-comment rulemaking is where firms ask for these instruments. Business comments move final rules more than other comments (Yackee & Yackee, 2006); firms that meet with regulators gain distributive benefits (Libgober, 2020a); larger and more diverse coalitions move rules further (Dwidar, 2022; Nelson & Yackee, 2012); and firms keep lobbying after a policy is enacted to shape how it is implemented (You, 2017). Agencies also anticipate comments when they draft, so observed changes understate influence on the draft itself (Libgober, 2020b); and comments that appear independent can carry their funders' interests (Bertrand et al., 2021).

If carve-outs are part of the price of expansion, they should arrive with it, and they should go to the firms whose resistance the state most needs to manage.

H2a. BIS revisions that widen control scope also add carve-outs.

H2b. Carve-outs go disproportionately to the coalition that opposes expansion most.

The second way is fiscal. Compensation has long sustained support for costly economic policy. Embedded liberalism paired openness with domestic insurance (Ruggie, 1982), open economies run larger governments (Rodrik, 1998), and compensation sustains public support for openness (Hays, 2009; Walter, 2010). Adjustment assistance and protection work as substitutes (S. E. Kim & Pelc, 2021), and assistance to those hurt by trade dampens

backlash (Ritchie & You, 2021). Issue linkage builds domestic support for costly agreements (Davis, 2004). In climate policy, industrial policy builds coalitions that later support costly regulation (Meckling et al., 2015), and targeted compensation raises support for transition among those it pays (Bolet et al., 2024; Gaikwad et al., 2022). The return of industrial policy (Evenett et al., 2024; Juhász et al., 2024) gives states a new set of instruments for this purpose, and chip subsidies, tariffs and controls form one policy package (Bown & Wang, 2024; Luo & Van Assche, 2023). Industrial policy also raises the question of who captures it (Breznitz & Gingrich, 2025; Juhász & Lane, 2024).

Fiscal compensation is competed for. The CHIPS Act left the allocation of its manufacturing incentives to the Department of Commerce, the department in which BIS also writes the export controls. Firms applied under a notice of funding opportunity. Commerce signed a preliminary memorandum of terms with applicants it judged “reasonably likely to receive an award,” reviewed their projects and then negotiated final terms (Congressional Research Service, 2026). At each step Commerce decided who would be paid, how much and on what conditions. That kind of discretion leaves room for lobbying to shape the outcome. When firms applied for capital under the Troubled Asset Relief Program, those with political connections, including lobbying and campaign contributions, were more likely to be funded (Duchin & Sosyura, 2012).

The firms that fight the controls are also well placed to win this competition. Lobbying on export controls shows how much a firm stands to lose, and a state that pays losers to keep a policy going has reason to direct its money toward them. The same lobbying gives a firm working access to the department that decides its award. Access brings firms

distributive benefits in rulemaking (Libgober, [2020a](#)), and a firm already in regular contact with Commerce over the scope of the controls can carry that relationship into the negotiation of its award. The payment is therefore partly a product of the firm’s political activity, which makes the step from lobbying to award a separate question from the step from award to behavior. Firms that lobby harder through the application process should be more likely to come away with an award.

H3a. Firms that lobby more heavily on export controls are more likely to secure a CHIPS award.

What payment changes depends on the terms the firm signs. A preliminary memorandum of terms does not guarantee an award (Congressional Research Service, [2026](#)). A firm that holds one still has money to win, and its lobbying on chips and export controls serves its case for the award as well as its case against the controls. Finalization turns the terms into a contract that binds the firm to the national security guardrails of the CHIPS Act. For ten years after the award, a recipient that materially expands semiconductor manufacturing capacity in China or another foreign country of concern can be made to repay the full amount, with exceptions for legacy chips (88 FR 65600, 65601). A firm that has accepted that limit has less at stake in the Chinese plants that the controls also reach. Commerce pays out awards as firms complete negotiated project milestones, and some projects are scheduled to run into the 2030s (Congressional Research Service, [2026](#)), so a paid firm depends on the department for years after its award is signed.

Payment acts mainly on how hard a firm fights. A paid firm can still believe the

controls cost it money. A firm that has received relief in the rule text has less left to ask for, though, and once its award is secured it has less to gain from pressing its case. Because its disbursements run through Commerce, it also has more to lose from a public fight with the department. Payment should therefore reduce a firm’s mobilization against the controls, whatever happens to its stated position.

H3b. Firms whose CHIPS award finalizes reduce their export-control lobbying relative to firms that reached preliminary terms and never finalized.

If the firms that resisted were paid and their resistance faded, the controls could keep widening without the losers setting their terms, and the state would keep the direction of policy while conceding its distribution.

3 Data and Measurement

3.1 The Comment Corpus

I collected every public comment on 49 dockets from Regulations.gov, 6,641 comments filed between 2022 and 2026 to 16 agencies. The core is the eight BIS dockets on the chip, advanced computing and AI rules described in Section 3.2 and the Commerce Department’s 2022 request for information on CHIPS incentives (DOC-2022-0001). The rest are other BIS dockets on export controls, on supply-chain rules for connected vehicles and drones and on Section 232 investigations of critical minerals and robotics, and every

non-BIS docket since 2022 with artificial intelligence in a document or docket title and at least one public comment, among them dockets at the National Telecommunications and Information Administration (NTIA), the Office of Science and Technology Policy (OSTP), the National Institute of Standards and Technology (NIST), the Office of Management and Budget (OMB), the Federal Trade Commission (FTC), the International Trade Administration (ITA) and the Patent and Trademark Office. Many firms file comments as PDF attachments, so the text of each comment includes its attachments, and letters that BIS posted as page images were recovered by optical character recognition. Comments under 500 characters (762) cannot state a position on scope and are dropped, which leaves 5,879 scored comments.

Campaign copies and coordinated letters are common. On the 2024 round of the advanced computing rule, letters from three industry groups share hundreds of identical twelve-word passages, including drafted regulatory text, and on the equipment docket Applied Materials, KLA and Lam Research filed recommendations from one template. I group comments into clusters that share at least half of their sampled twelve-word strings or come from the same organization, 5,570 clusters in all, and cluster standard errors on them.

3.2 Rules and Awards, 2022 to 2026

Table 1 lists the BIS rules whose dockets generate the corpus. The October 2022 interim final rule did four things. It created license requirements for advanced computing chips (ECCN 3A090) and computers containing them (4A090) destined for China; controlled

equipment for producing advanced-node chips; extended end-use controls to supercomputer and semiconductor manufacturing in China; and required licenses for US persons supporting advanced-node production there. BIS issued it as an interim final rule, effective before comment, and invited comments afterward.

Table 1. BIS semiconductor and advanced computing rules with a public comment record, 2022 to 2025

Effective	FR document	Citation	Rule	Comments
2022-10-07	2022-21658	87 FR 62186	Advanced computing and SME controls	49
2023-11-17	2023-23049	88 FR 73424	SME rule	23
2023-11-17	2023-23055	88 FR 73458	Advanced computing update	41
2024-03-13	2024-05267	89 FR 18353	Radiation-hardened ICs, License Exception GOV	5
2024-04-04	2024-07004	89 FR 23876	Corrections and clarifications	15
2024-12-02	2024-28270	89 FR 96790	FDP additions, HBM and SME refinements	13
2025-01-13	2025-00636	90 FR 4544	AI diffusion framework (not enforced)	67
2025-01-16	2025-00711	90 FR 5298	Foundry due diligence	15
2025-09-29	2025-19001	90 FR 47201	Affiliates rule	29
2025-11-10	2025-19846	90 FR 50857	One-year suspension of the affiliates rule	0

Note. Effective dates and comment counts per Federal Register document, from the Federal Register API. SME is semiconductor manufacturing equipment, FDP the foreign direct product rule and HBM high bandwidth memory. Later corrections (2023-27588, 2025-02655) and the April 2026 extension of the authorized IC designer date (2026-06851) drew no comments and are omitted.

Revisions came in waves. In October 2023 BIS split the rule in two, one on advanced computing and one on equipment, each responding at length to the comments received. The computing rule redefined the chip thresholds to capture chips designed to fall just below the 2022 line and added a notification-based license exception. The equipment rule folded the 2022 equipment controls into existing ECCNs, extended license requirements to all arms-embargoed destinations (Country Group D:5) and narrowed several provisions that commenters had called overbroad. A corrections rule followed in April 2024. In December 2024 BIS added new controls on equipment, two foreign direct product rules that reach equipment made abroad with US technology, and controls on high bandwidth memory. In

January 2025 it issued the AI diffusion framework, which Commerce announced on May 13, 2025, two days before its compliance date, that it would not enforce and would rescind (U.S. Government Accountability Office, 2026), and a foundry due diligence rule that requires foundries and packaging firms to presume that certain advanced chips are controlled unless the designer or packager is on an approved list. In September 2025 BIS extended Entity List restrictions to any firm at least 50 percent owned by a listed entity. Six weeks later it suspended that affiliates rule for a year.

The CHIPS and Science Act, signed on August 9, 2022, appropriated \$39 billion for manufacturing incentives through fiscal year 2026 (Congressional Research Service, 2026). The CHIPS Program Office announced preliminary memoranda of terms from December 2023 through January 2025 and finalized most awards between November 2024 and January 2025. Twelve firms received preliminary terms and no final award. Figure 3 plots the dates.

After January 2025 the channel changed. The White House announced the one-year suspension of the affiliates rule on November 1, 2025, as part of a trade arrangement with China (The White House, 2025), and BIS implemented it on November 10 by a final rule without notice and comment.¹ In January 2026 BIS moved license review for Nvidia’s H200 and less advanced chips to China from a presumption of denial to case-by-case review, again by a final rule issued without prior comment (91 FR 1684). Decisions on chip sales moved from the rulemaking record to the White House. Section 7 returns to this shift.

¹The suspension rule itself (90 FR 50857) gives no reason beyond continued evaluation of national security and foreign policy interests.

3.3 Stance on Control Scope

The outcome for H1 is each comment’s stance on control scope, from -2 (asks for permanent narrowing, such as an exclusion, a raised threshold or withdrawal) through 0 (no net position) to $+2$ (asks that more be controlled). On the CHIPS docket the same scale measures stance on the conditions attached to CHIPS money. On the other dockets it measures whether the commenter wants the rule or the policy to control more or less.

I hand coded 213 letters, the 182 eligible comments on the eight BIS chip dockets and a stratified sample of 31 from the CHIPS docket, under a written codebook (Appendix A.3). Each letter is coded on its own text, so a filer’s other letters do not carry a code across. The hand codes serve as training data and validation, and every test uses model scores.

The scorer is a DeBERTa-v3 natural language inference model (Laurer et al., 2024) fine-tuned on the hand codes. Each comment is split into windows of 1,800 characters, and the model judges each window against four statements (the commenter wants the rule to control less, supports it or wants more, asks for specific narrowing, asks for specific widening) whose probabilities combine into the five-point scale. A second head assigns one of five argument types, which record where the commenter argues from (upstream inputs, midstream equipment and fabrication, downstream deployment, compliance process, or national security). Training is cross-fitted in five folds, with folds grouped by template cluster so that copies of one letter never sit on both sides of a split, and gold comments are scored by the model that did not see them.

Out of fold, the scorer’s stance correlates with the hand codes at 0.78 (Spearman) across all 213 letters and at 0.73 on the BIS letters, against 0.64 for the same model before fine-tuning (Table A1). Argument type matches the hand code for 78 percent of letters. Validation is weaker on the CHIPS docket (0.49) and absent on the AI dockets, which have no hand codes. I therefore read the pooled results through specifications that compare commenters within agency or within docket. Table 2 shows five letters with their hand codes and out-of-fold model scores.

3.4 Position on the Chain

H1 needs each commenter’s stage on the AI compute chain. I code it from what the organization does, the same in every docket. Upstream covers chip design, IP cores, design software and research. Midstream covers equipment, materials, fabrication, packaging and test. Downstream covers everyone who buys or uses chips or AI. Diversified firms take their largest business, law firms and investors take the client they file for, and small firms with little public information default to downstream as users of AI. Unions and universities have no product, so each of their comments takes the stage of the workers or research it speaks for. Associations, think tanks, individuals and governments have no stage and stay out of H1. In total, 1,657 organization names were coded, and the H1 sample is 1,213 comments from firms, unions and universities in 944 clusters and 36 dockets (1,052 downstream, 126 midstream and 35 upstream). A coding relative to each rule’s controlled item, done by hand for the semiconductor dockets, is kept for robustness.

Table 2. Five comment letters, hand codes and model scores

Commenter	Chain	Excerpt	Hand code	Model score
KLA BIS-2023-0016	Midstream	“Several of the comments are focused on ensuring a level regulatory playing field between U.S. semiconductor manufacturing equipment (‘SME’) companies and their competitors in the European Union, Japan, South Korea, and other allied countries.”	−2	−1.97
Cerebras BIS-2022-0025	Upstream	“We respectfully suggest that it is important to support U.S. commercial innovation and leadership by allowing companies to continue to recruit and retain the best and brightest talent, and avoiding overly broad restrictions on the nationalities of available talent.”	−1	−1.80
Lenovo BIS-2022-0025	Downstream	“[A]ny definition adopted by BIS should exempt organizations [...] with an ultimate parent located in Macau or Country Group D:5, but the day-to-day operations of which are not controlled by an entity located in Macau or Country Group D:5.”	−2	−1.51
MathWorks BIS-2022-0025	Downstream	“[W]here BIS must impose restrictions, it should do it even handedly across all distribution models.”	+2	0.35
Integra Technologies DOC-2022-0001	Midstream	“We believe companies that currently operate in China or expand in China that utilize any of these packaging techniques should be subject to clawback provisions.”	+2	0.38

Note. Excerpts are verbatim. Hand codes run from −2 (asks for permanent narrowing) to +2 (asks that more be controlled). Model scores are continuous and come from the fold that did not train on the letter. KLA asks for relief from US-person controls that its allied competitors do not face. Cerebras asks BIS to keep deemed exports outside the license requirement. Lenovo asks for an exemption from the headquarters test. MathWorks asks BIS to hold cloud delivery of software to the end-user controls that apply to its downloadable products. On the CHIPS docket, Integra asks that the expansion clawback cover advanced packaging.

3.5 Rule Transitions and Carve-outs

For H2 the unit is a transition from a commented rule to its next version. Seven transitions qualify (Table 4). Each is coded for whether the revision widened, held or narrowed control scope, with corrections that widen in substance counted as widening, and for whether it added any exemption. Within the seven revisions I coded every carve-out, 57 in all, meaning every provision that relieves some party from a control that would otherwise apply. Each is coded for the stage of the chain it relieves (one or more of upstream, mid-stream and downstream, or general when it relieves all exporters or the government) and for whether BIS’s own preamble says it granted the relief in response to comments. Each code rests on a quotation from the Federal Register text.

3.6 Lobbying and Awards

For H3 I build a firm-quarter panel from 2021 to 2025 out of 12,996 LD-2 lobbying disclosures. A filing counts toward a firm’s chip and export-control lobbying when its activity text mentions semiconductors, export controls, chips, the CHIPS Act, the Entity List, advanced computing, foundries, wafers, lithography, microelectronics, the Export Administration Regulations, ECCNs or artificial intelligence, or when it lists BIS among the agencies lobbied. CHIPS preliminary and final award dates come from Congressional Research Service (CRS) Report R49031 (Congressional Research Service, 2026). The panel covers 78 firms and associations, with supply-chain position coded by hand.

4 Who Resists

Table 3 reports comment-level regressions of the scope score on chain position, with midstream as the reference. Agency fixed effects are the main specification, since stance means different things across agencies and the CHIPS and AI dockets have their own baselines. Docket fixed effects compare commenters on the same rule. Because 36 dockets give few clusters for a docket-level test, I report wild cluster restricted bootstrap p-values at the comment-cluster level and by docket.

Commenters off the midstream ask for wider control, or for less narrowing, than midstream commenters. With agency fixed effects the gap is 0.26 points on the five-point scale (bootstrap $p = 0.02$ by comment cluster). Holding argument type fixed leaves it at 0.25, so the gap holds among commenters who make the same kind of argument. With docket fixed effects, which compare firms commenting on the same rule, it is 0.31 ($p = 0.002$ by cluster and 0.03 by docket). Restricting the sample to firms gives 0.28 and 0.34 (Appendix Table A2). The docket-level bootstrap is weaker under agency fixed effects (p between 0.15 and 0.22), which reflects how few dockets have commenters from both groups.

Downstream firms drive the gap (0.28 in column 2). Upstream firms are close to midstream firms (0.06, standard error 0.12), but only 35 upstream comments enter the sample and nearly all of them come from outside the chip rules. Chip designers are close to absent from the BIS record. On the chip rules two upstream firms, Cerebras and AMD, filed one comment each, against 24 comments from 14 midstream firms.

Table 3. Chain position and stance on control scope

Dependent variable	Scope score			
	(1)	(2)	(3)	(4)
Upstream or downstream	0.257*** (0.097) [0.020] {0.221}		0.253*** (0.092) [0.010] {0.146}	0.314*** (0.091) [0.002] {0.030}
Downstream		0.280*** (0.099) [0.015] {0.197}		
Upstream		0.057 (0.121) [0.646] {0.724}		
Agency fixed effects	Yes	Yes	Yes	No
Argument-type fixed effects	No	No	Yes	No
Docket fixed effects	No	No	No	Yes
Observations	1,212	1,212	1,212	1,211
Comment clusters	943	943	943	943
R ²	0.491	0.492	0.555	0.534
Within R ²	0.008	0.010	0.006	0.009

Note. OLS estimates. The dependent variable is the fine-tuned scope score (−2 narrow to +2 wide), and midstream commenters (equipment, materials, fabrication, packaging and test) are the reference group. The sample is firms, unions and universities. Standard errors clustered on comment clusters (campaign copies and repeat filers) in parentheses. Wild cluster restricted bootstrap p -values (Webb weights, 999 draws) in brackets at the comment-cluster level and in braces by docket. The affiliates rule docket (BIS-2025-0017) has no controlled item and is excluded. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$ from the clustered standard errors.

The gap shows up rule by rule where both groups comment. Off-midstream commenters score 0.66 points above midstream commenters on the October 2022 rule (9 against 11 comments), 0.63 on the equipment rule, 0.34 on the CHIPS docket (46 against 72) and 0.65 on ITA’s 2025 docket on an AI exports program. The one exception is the Section 232 robotics docket, where two midstream comments sit above 134 others. On the BIS chip rules every group asks for narrowing on average, and midstream coalitions ask for more of it than downstream coalitions on every rule where both comment (Table 4).

5 Expand and Exempt

Table 4 lists the seven transitions with the mean stance of each coalition on the rule that preceded the revision. On every rule every coalition with at least one member asked for narrower control. Five of the six chip-rule revisions widened scope anyway. The sixth (T7, from the foundry rule to its correction and the April 2026 extension of the designer deadline) held scope flat. The seventh transition, the suspension of the affiliates rule, is the only one that narrowed, and it came by a final rule without comment. Every one of the seven revisions added at least one carve-out.

The pattern contradicts the view that firms set the direction of the controls. No coalition asked for wider control on any of these rules, and the coalitions differed only in how hard they pushed for narrowing.

Table 5 shows where the carve-outs landed. 34 of the 57 relieve midstream firms, 28

Table 4. Rule transitions, coalition stance and scope change

Transition	From	To	Scope	Mean scope score (organizations)		
				Upstream	Midstream	Downstream
T1	2022-21658	2023-23055, 2023-23049	Wider	-1.86 (1)	-1.40 (10)	-1.38 (7)
T2	2023-23055	2024-07004	Wider	-1.61 (3)	-1.83 (3)	-1.02 (9)
T3	2023-23049	2023-27588, 2024-07004	Wider	-	-1.39 (6)	-
T5	2024-07004	2024-28270	Wider	-	-1.10 (4)	-0.74 (2)
T6	2024-28270	2025-00711	Wider	-	-1.78 (4)	-0.30 (2)
T7	2025-00711	2025-02655, 2026-06851	Flat	-1.97 (1)	-1.44 (3)	-1.42 (2)
T9	2025-19001	2025-19846	Suspended	-1.08 (2)	-1.38 (3)	-1.24 (7)

Note. Federal Register document numbers. Mean fine-tuned scope score of the organizations at each stage commenting on the rule a transition starts from, with the number of organizations in parentheses. Coalitions include firms, unions, universities and associations by membership. Scope is the coded change from the earlier rule to the revision. Every revision added at least one carve-out. T4 (a comment period extension), T8 (no successor rule) and T10 (the AI diffusion framework, set aside by a May 2025 announcement of non-enforcement with no rescission in the Federal Register) are dropped.

of them midstream firms alone. Fourteen relieve upstream firms and 13 relieve downstream firms, mostly in exceptions shared with another stage. Ten are general, such as savings clauses for shipments en route and exceptions for US government use. BIS says that 21 of the 57 respond to comments, and 15 of those 21 reach midstream firms. In the same rules, midstream items account for about a third of the controlled ECCN provisions (110 of 329). Relief went to midstream firms at roughly twice their share of what was controlled.

The carve-outs also differ in what they let through. I coded each one into three groups. The first lets items, technology or US-person services reach end users in China without a license. The second relieves operations outside China, allied staff or firms' own supply chains, and the third settles timing or administration. 24 of the 57 reopen flows to China in this sense, and 18 of those 24 reach midstream firms (Table 5). All 8 requested carve-outs that reopen flows to China reach midstream firms. About half of the relief for midstream firms reopened flows to China (18 of 34), against 4 of 14 carve-outs for upstream firms and 4 of 13 for downstream firms. Resolving every close call one way or the other

Table 5. Carve-outs in seven BIS revisions by the stage of the chain they relieve

	All carve-outs	Granted at commenters' request	Reopens flows to China
Reaches midstream firms	34	15	18
midstream alone	28	13	18
Reaches upstream firms	14	4	4
Reaches downstream firms	13	3	4
General (all exporters or the government)	10	2	1
Total	57	21	24

Note. A carve-out relieves some party from a control that would otherwise apply. A carve-out can reach more than one stage, so the stage rows overlap. “Granted at commenters’ request” means BIS’s preamble attributes the provision to public comments. Nine carve-outs soften a control created in the same revision, and they are included. “Reopens flows to China” counts carve-outs that let items, technology or US-person services reach end users in China, Macau or Country Group D:5 without a license. Items sent into China for US or allied firms’ own supply chains are not counted, and allied firms’ reexports to China are. Six of the 24 are close calls, and five carve-outs coded otherwise are close calls the other way.

leaves midstream firms with between 72 and 83 percent of the carve-outs that reopen flows to China.

The carve-outs that BIS granted on request show the mechanism. In the October 2023 equipment rule BIS narrowed its end-use control on equipment for Chinese toolmakers from any item subject to the EAR to items on the Commerce Control List, writing that “BIS agrees that this provision is overbroad.” It exempted back-end assembly, test and packaging from the definition of production, excluded masks and reticles, gave US and allied equipment makers a temporary general license to keep using suppliers in China, and excluded US-person employees of allied fabs and equipment firms from the US-person controls. It dropped a control on cobalt deposition tools, saying that “based on feedback from industry, including from this commenter, BIS agrees” that the control was poorly tailored. In one case the rule text tracks a comment closely. An anonymous commenter on the 2022 rule drafted replacement text for the control on atomic layer deposition tools, and BIS adopted nearly the

same four-condition structure in 2023 while setting a different temperature threshold. The January 2025 foundry rule, which widened the presumption of control over chips made by foundries, was issued “in response to requests from the public” for due diligence procedures, and its approved designer and approved packager lists let foundries and packaging firms certify their way out of the presumption.

In each case the rule that granted midstream relief also extended the controls over the same firms’ products and plants.

6 Paying the Resisters

CHIPS money went to the coalition that resisted most. Of \$33.5 billion in incentive awards that reached preliminary or final terms, \$32.8 billion went to 26 midstream firms, the foundries, memory makers, integrated manufacturers, equipment makers and materials suppliers.² No upstream design firm received one.

6.1 From Lobbying to Award

H3a asks whether lobbying predicts receiving preliminary terms. Table 6 reports a complementary log-log hazard of preliminary terms by quarter, with the risk set running from the first to the last quarter in which any firm received terms, calendar-quarter effects, size and supply-chain position controls, and standard errors clustered by firm. Associations

²Downstream recipients by main business are HP, Infinera, Rocket Lab, Corning, BAE Systems and Bosch, about \$0.6 billion together. Arizona State University and Activate account for the rest.

are dropped. The sample has 68 firms, 357 firm-quarters and 26 events.

Table 6. Lobbying and the hazard of receiving CHIPS preliminary terms

Dependent variable	Preliminary terms			
	(1)	(2)	(3)	(4)
Filings, previous quarter (log)	0.318 (0.302)			
Filings, previous four quarters (log)		0.389 (0.259)		
Filings, cumulative to date (log)			0.362 (0.299)	
Spend, previous four quarters (log)				0.093 (0.063)
Firm size and supply-chain position	Yes	Yes	Yes	Yes
Calendar-quarter fixed effects	Yes	Yes	Yes	Yes
Firm-quarters	357	357	357	357
Preliminary terms (events)	26	26	26	26

Note. Complementary log-log discrete-time hazard of receiving CHIPS preliminary terms, one model per lobbying measure, 68 firms. Coefficients are log hazard ratios, so $\exp(\beta)$ gives the hazard ratio. Filings count LD-2 reports on chips and export controls, and filings and spend enter as $\log(1+x)$. Standard errors clustered by firm in parentheses. $*p < 0.1$, $**p < 0.05$, $***p < 0.01$.

Every lobbying measure points in the expected direction, and none is statistically distinguishable from zero. A one-unit increase in the log of a firm's filings over the previous four quarters multiplies the hazard by 1.47 (95 percent interval 0.89 to 2.45). With 26 events the test has little power. An event study of the same panel, comparing firms that received preliminary terms with firms that never did, shows lobbying rising before terms were announced, with estimates of 0.12 to 0.17 two and three quarters before and 90 percent intervals that exclude zero at two quarters (Figure 1). The award appears to be partly a product of lobbying, which is why the next step compares only firms that had already received terms.

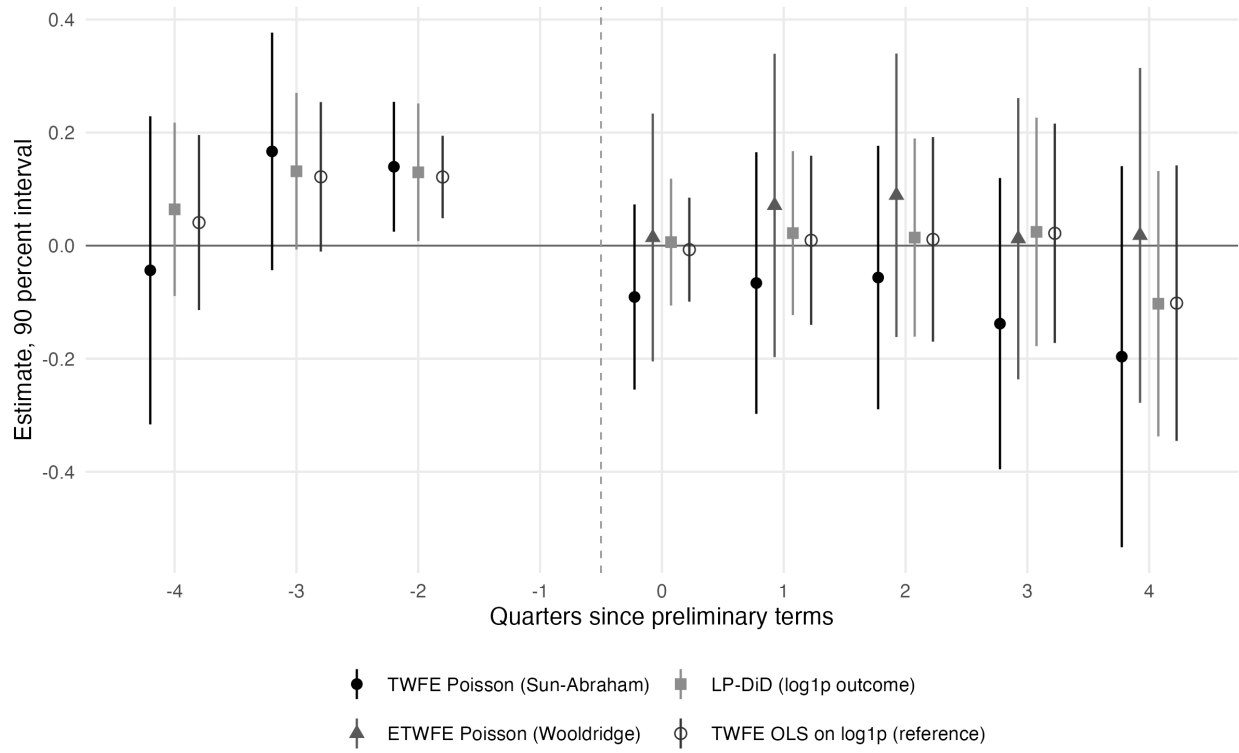


Figure 1. Chip and export-control lobbying around CHIPS preliminary terms, firms that received terms against firms that never did. 77 firms, 25 with terms. Points are event-time estimates with 90 percent intervals. Event -1 is the reference. Pre-period estimates describe the H3a mechanism and are not a test of parallel trends.

6.2 From Award to Demobilization

H3b compares the 16 firms in the panel whose awards were finalized with the 10 that received preliminary terms and never finalized. Holding preliminary terms fixed holds selection into the program fixed, so the pre-period coefficients carry their usual meaning as a test of parallel trends. Figure 2 plots event-time estimates from four estimators (Dube et al., 2025; Sun & Abraham, 2021; Wooldridge, 2023).

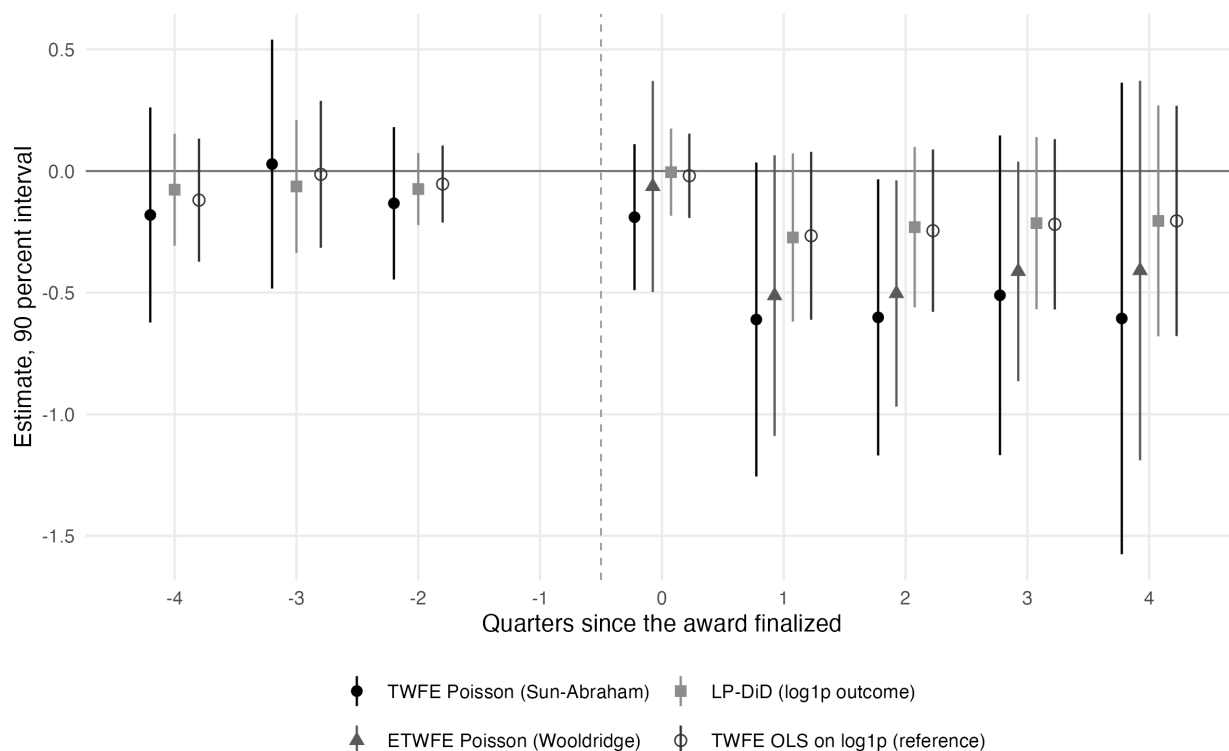


Figure 2. Chip and export-control lobbying after CHIPS award finalization, finalized firms against firms with preliminary terms only. Outcome is LD-2 filings per firm-quarter. Points are event-time estimates with 90 percent intervals. 26 firms, 16 finalized. Event -1 is the reference.

Before finalization the two groups move together. All pre-period estimates are small and none is distinguishable from zero. After finalization, finalized firms file less. In the Sun and Abraham Poisson model filings fall by about 0.6 log points one and two quarters

after the award (-0.61 , standard error 0.39 , and -0.60 , standard error 0.35) and stay down through the fourth quarter. The Wooldridge Poisson estimates are similar (-0.50 in the second quarter, 90 percent interval -0.97 to -0.04). The log-linear estimators give smaller declines of about 0.25 that are not distinguishable from zero. Poisson weights by expected counts, and Micron, Intel and Samsung account for much of the filing volume, so the decline is concentrated among the largest filers. The design is thin. Final awards fall in two quarters, 2024Q4 and 2025Q1, and the panel ends in 2025Q4.

6.3 Timing

Figure 3 lays the award dates over the rule calendar. 20 of the 21 final awards came between November 15, 2024 and January 17, 2025. The December 2, 2024 rule, which added equipment controls and foreign direct product rules aimed at equipment makers, took effect within three weeks after TSMC, GlobalFoundries and Intel finalized their awards, and Micron, SK hynix, Samsung and Texas Instruments finalized theirs in the three weeks that followed. The January 16, 2025 foundry rule took effect the same week that Applied Materials, Corning, Edwards Vacuum and Infinera finalized theirs.

7 Statecraft or Private Interest

US chip export controls are security statecraft that private interest hollows out from within. The state set the direction of the controls. Every coalition that commented asked

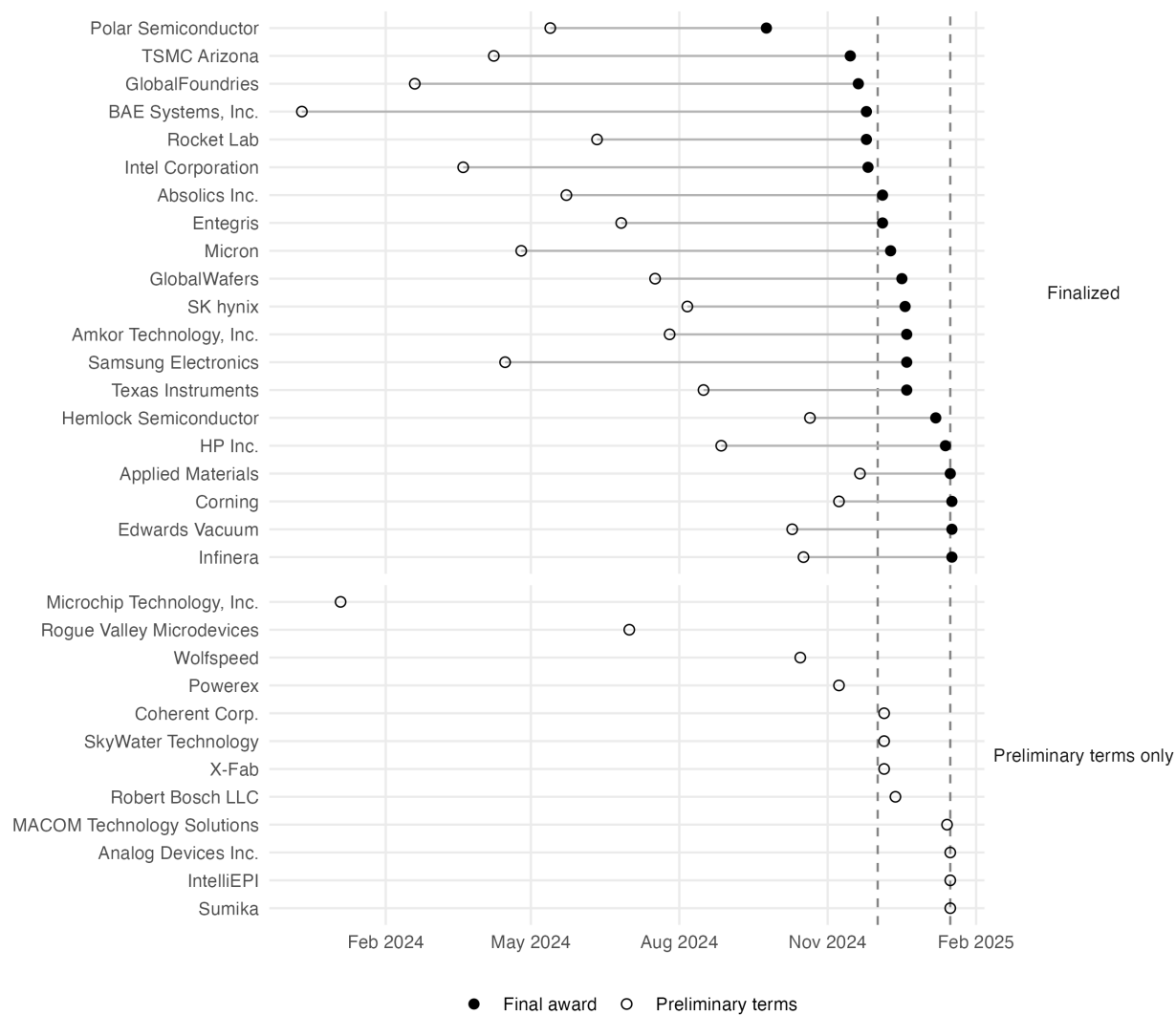


Figure 3. CHIPS preliminary terms and final awards by firm. Dashed lines mark the December 2, 2024 rule on equipment, foreign direct product rules and high bandwidth memory, and the January 16, 2025 foundry due diligence rule. Dates from CRS Report R49031.

for less control on every chip rule, including the downstream coalition that the commercial reading expects to gain from wider controls, and BIS widened scope in five of six revisions anyway. The firms that resisted hardest decided where the wider controls would bite. Relief went to midstream firms at roughly twice their share of the controlled provisions, and 15 of the 21 carve-outs that BIS granted at commenters' request reached them.

Exemptions for midstream firms carry a security cost because those firms hold a chokepoint. A state's leverage in an interdependent network rests on the hubs it controls (Farrell & Newman, 2019), and advanced equipment is one of the US hubs in chipmaking. 18 of the 24 carve-outs that reopened flows to China went to midstream firms, including all 8 that BIS granted at commenters' request. The October 2023 equipment rule narrowed the end-use control on Chinese toolmakers from any item subject to the EAR to items on the Commerce Control List, so uncontrolled US inputs could again reach Chinese firms that build chipmaking equipment and compete with US toolmakers. The same rule dropped a control on cobalt deposition tools. Outside the revisions coded here, Validated End-User authorizations let suppliers ship controlled items to Samsung's and SK hynix's memory plants in China without licenses from October 2023 until the end of 2025 (88 FR 71478). Most of the remaining midstream relief eased allied operations or firms' own supply chains. The carve-outs that reopened flows to China gave up part of the leverage the controls were built on, and most of them went to the firms that hold that leverage.

Chip designers show the same pattern outside the rulemaking record. They lose most from controls on chip sales, filed almost no comments and could not be paid through manufacturing incentives. They had already built chips to fall just below the 2022 thresholds,

which BIS redefined in October 2023 to capture them. Their exemption came from the White House. In August 2025 the President said he had agreed to approve Nvidia's sales of its H20 chip to China in return for 15 percent of the revenue (Trump, [2025a](#)). On December 8, 2025 he announced that Nvidia could ship its H200 to approved customers in China with 25 percent paid to the United States (Trump, [2025b](#)), and in January 2026 BIS moved H200 license review to case-by-case by a final rule issued without comment (Bureau of Industry and Security, [2026](#)). Here the exempted item was the advanced AI chip itself, the item the October 2022 rule was first written to keep from China. These decisions came under a new administration and alongside a trade arrangement with China (The White House, [2025](#)), which may explain their timing. They still followed the same line, with relief going to the firm whose sales the controls had cut.

The CHIPS money worked differently from the carve-outs. Final awards bound recipients to the ten-year Expansion Clawback, so the payment carried conditions that served the security aim, and the paid firms lobbied less on chips and export controls once their awards were secure. The state paid its strongest resisters twice, in the rule text and in cash, and the cash came with security conditions attached.

Expand and exempt means the scope written into the rules overstates how much the controls restrict China, because the exemptions sit where US leverage is greatest. The security label also shelters the bargain. Calling an issue a security matter lifts it out of normal politics (Buzan et al., [1998](#)), and the exemptions were negotiated inside that frame, first in BIS's responses to comments and later in announcements by the President.

8 Conclusion and Future Work

US chip export controls are security statecraft that private interest hollows out from within. They widened while every organized coalition in the record asked for less, and the firms that resisted hardest decided where the wider controls would bite. Of the 24 carve-outs that reopened flows to China, 18 went to midstream firms, including all 8 that BIS granted at commenters' request. These firms make the equipment and materials and run the fabs that give the United States its hold over Chinese chipmaking, so the relief they won came out of the leverage the controls were meant to use.

Counts of controlled items and listed entities therefore overstate how far the United States has restricted China, because the exemptions are concentrated in chipmaking equipment and services, where US leverage is greatest. Assessments of the controls need to read the carve-outs alongside the expansions. The form of payment matters as well. CHIPS awards came with a ten-year bar on expanding capacity in China, so the cash bought quiet on terms that served the security aim, while the carve-outs bought quiet by reopening flows to China. If the state wants wider controls to keep their bite, paying the losers in cash with conditions costs less security than paying them in exemptions.

For the study of economic statecraft, the case shows a limit on weaponized interdependence that comes from home (Farrell & Newman, [2019](#)). The firms that own a chokepoint give the state its leverage and lose the most when the state uses it, so their resistance is aimed at the source of that leverage. The more a state relies on chokepoints held by its own firms, the more its controls depend on how it pays those firms.

Chip designers are the gap in this record. They lose most from controls on chip sales, are nearly absent from the comment record and cannot be paid through manufacturing incentives. Their exemptions came from the White House after 2025, in sales approvals for Nvidia’s H20 and H200 that traded access to China for a share of revenue. Earnings call transcripts for US-listed firms across the chain would bring designers into the H1 comparison, give H3b a stance measure for firms with preliminary terms only, and address selection into commenting (Hassan et al., [2019](#), [2024](#)).

Validation of the scorer is strong on the BIS dockets, weaker on the CHIPS docket and absent on the AI dockets that supply most of the H1 sample. The lobbying measure counts filings that mention chips, export controls or the CHIPS Act, so part of the post-award decline may reflect firms that stopped lobbying for an award they had won. A count restricted to export-control activity would separate the two. The H3b comparison rests on 26 firms and two award quarters. The carve-out coding counts provisions and treats each one as a unit regardless of the trade it covers, and eleven of the flow codes are close calls. Weighting carve-outs by trade would show how much of the controls’ reach the exemptions gave back.

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A. Appendix

A.1 Scorer Validation

Table A1. Out-of-fold validation of the fine-tuned scorer against hand codes

	Letters	Spearman	Scope F1	Argument type
All hand-coded letters	213	0.78	0.61	0.78
BIS dockets	182	0.73	0.60	0.81
CHIPS docket (DOC-2022-0001)	31	0.49	0.39	0.61
BIS dockets, base model	182	0.64	0.53	0.50

Note. Scores for each letter come from the fold that did not train on it. Scope F1 is weighted and cuts the continuous score at the hand-coded label shares. Argument type is exact agreement. The base model is the same NLI model before fine-tuning. The pooled Spearman is inflated by the level gap between BIS and CHIPS letters. Argument type recall and precision are 0.89 and 0.85 for midstream equipment, 0.64 and 0.58 for upstream inputs and 0.52 and 0.54 for downstream deployment.

A.2 Firms Only

Table A2. Chain position and stance on control scope, firms only

Dependent variable	Scope score			
	(1)	(2)	(3)	(4)
Upstream or downstream	0.285*** (0.097) [0.010] {0.209}		0.285*** (0.088) [0.004] {0.175}	0.345*** (0.091) [< 0.001] {0.032}
Downstream		0.303*** (0.098) [0.005] {0.202}		
Upstream		0.110 (0.122) [0.372] {0.578}		
Agency fixed effects	Yes	Yes	Yes	No
Argument-type fixed effects	No	No	Yes	No
Docket fixed effects	No	No	No	Yes
Observations	1,152	1,152	1,152	1,151
Comment clusters	900	900	900	900
R ²	0.498	0.499	0.566	0.542
Within R ²	0.010	0.012	0.008	0.012

Note. Same specifications as Table 3, restricted to firms. Standard errors clustered on comment clusters in parentheses, wild cluster restricted bootstrap p -values in brackets (comment clusters) and braces (dockets). * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

A.3 Codebook Summary

Scope is coded on one five-point scale. -2 requires a request for permanent narrowing of something the rule already covers, such as an exclusion, a carve-out, a raised threshold, a rollback or withdrawal, and drafted replacement text that does this counts. Rejecting the premise of controlling at all is also -2 . -1 covers transitional relief, clarification requests that lean narrow, resistance to controls that do not yet exist and conditional narrowing. 0

means no net position, including clarification requests with no preferred answer. +1 endorses the rule as written or proposes drafting that leans wider without pressing for more capture. +2 asks that more be captured. Scope records direction only. A firm that asks for controls on its rivals' delivery channel is coded as widening. When asks conflict, the main argument sets the sign and the effect on the target country decides close cases.

Argument type records where the commenter argues from, regardless of side. Clarification letters are coded by the locus of the activity being defended, and narrowing letters by the locus of their narrowing asks, with the letter's emphasis taking precedence over a count of routine asks. Compliance process is reserved for letters whose grievance is the licensing machinery itself, and national security for letters whose end is security. On the CHIPS docket, scope is stance on the restrictions and conditions attached to CHIPS money, from asking to remove a statutory condition (−2) to asking to add or extend one (+2).